Computer Networks (EE-347)

Assignment 1: Client/Server Application using Sockets

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Server code (server.py)

```
import socket
from datetime import datetime

PORT = 6000

srv = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
srv.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1)
srv.bind(("", PORT))
srv.listen(5)
print(f"Server running on port {PORT}")

requests = 0
while True:
    conn, addr = srv.accept()
    try:
        name, roll = conn.recv(1024).decode().split(";", 1)
        requests += 1
        stamp = datetime.now().strftime("%Y-%m-%d %H:%M:%S")
        msg = (f"Hello {name} ({roll})\n"
               f"Your address: {addr[0]}:{addr[1]}\n"
               f"Request number: {requests}\n"
               f"Server date/time: {stamp}")
        conn.sendall(msg.encode())
        print(f"[{stamp}] Request {requests}: {name} ({roll}) from {addr[0]}:
{addr[1]}")
    except (ValueError, OSError) as e:
        print(f"Invalid request from {addr[0]}:{addr[1]}: {e}")
    finally:
        conn.close()
```

Client code (client.py)

```
import socket
import sys

HOST = sys.argv[1] if len(sys.argv) > 1 else "localhost"
PORT = 6000

name = input("Enter name: ")
roll = input("Enter roll number: ")

try:
    sock = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    sock.settimeout(10)
    sock.connect((HOST, PORT))
    sock.sendall(f"{name};{roll}".encode())
    print("\nServer says:\n" + sock.recv(1024).decode())
```

```
    sock.close()
except OSError as e:
    print(f"Connection to {HOST}:{PORT} failed: {e}")
```

friend.py

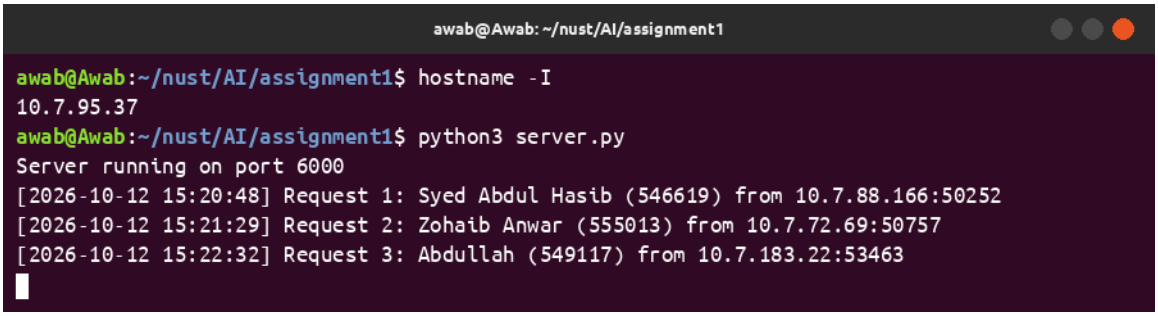
```
import runpy
import socket
import sys

ip, port = sys.argv.pop(1), int(sys.argv.pop(1))
_connect = socket.socket.connect

def connect(self, addr):
    self.bind((ip, port))
    return _connect(self, addr)

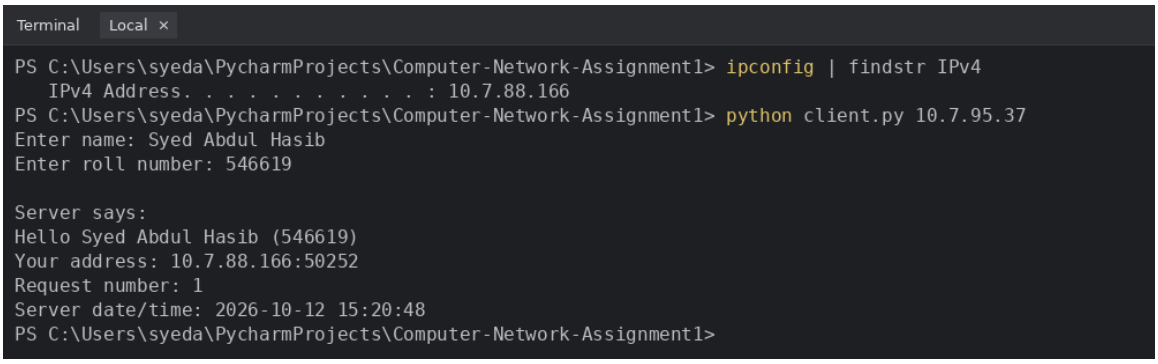
socket.socket.connect = connect
runpy.run_path("client.py", run_name="__main__")
```

Server output

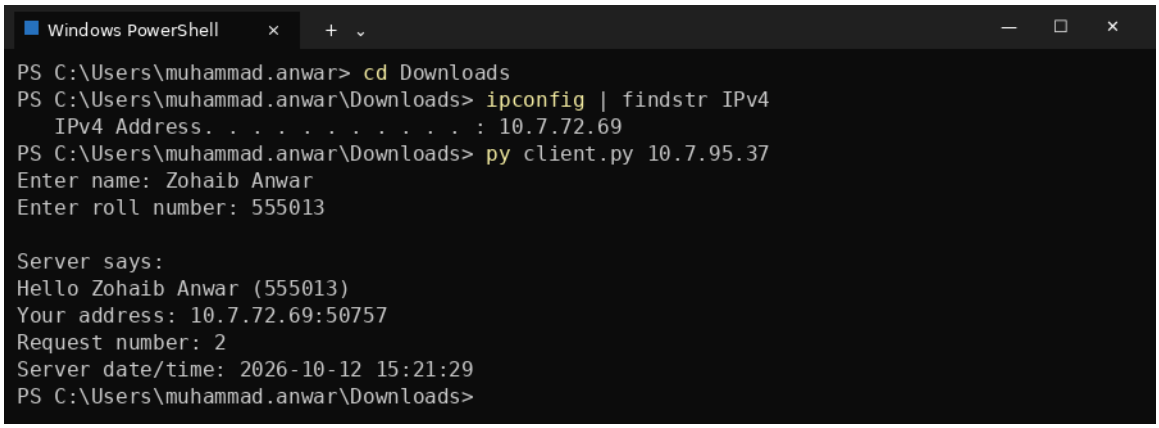


Client outputs

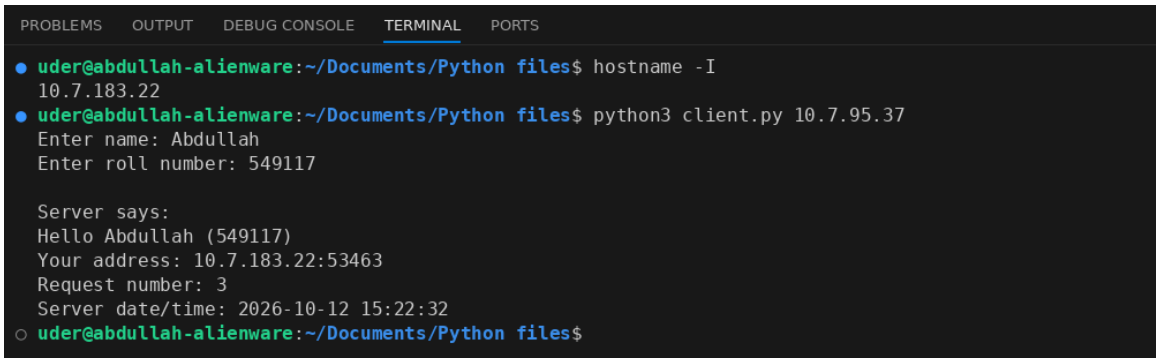
Hasib



Zohaib


Windows PowerShell
PS C:\Users\muhammad.anwar> cd Downloads
PS C:\Users\muhammad.anwar\Downloads> ipconfig | findstr IPv4
IPv4 Address. . . . . : 10.7.72.69
PS C:\Users\muhammad.anwar\Downloads> py client.py 10.7.95.37
Enter name: Zohaib Anwar
Enter roll number: 555013
Server says:
Hello Zohaib Anwar (555013)
Your address: 10.7.72.69:50757
Request number: 2
Server date/time: 2026-10-12 15:21:29
PS C:\Users\muhammad.anwar\Downloads>

Abdullah


PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
uder@abdullah-alienware:~/Documents/Python files$ hostname -I
10.7.183.22
uder@abdullah-alienware:~/Documents/Python files$ python3 client.py 10.7.95.37
Enter name: Abdullah
Enter roll number: 549117
Server says:
Hello Abdullah (549117)
Your address: 10.7.183.22:53463
Request number: 3
Server date/time: 2026-10-12 15:22:32
uder@abdullah-alienware:~/Documents/Python files$